EXPERIMENT 5

DATA VISUALIZATION USING MATPLOTLIB

AIM

To create and interpret different types of data visualizations using Matplotlib and Pandas for the social media dataset.

THEORY

Data visualization represents information graphically so that comparisons, distributions and relationships can be understood quickly. Bar charts compare categories, pie charts show proportions, histograms show numeric distributions, and scatter plots help examine relationships between two numeric variables. Titles, axis labels and readable scales are essential for correct interpretation.

ALGORITHM / PROCEDURE

1. Import Pandas and Matplotlib.
2. Load the dataset and select the columns required for each chart.
3. Count platform records and plot a bar chart.
4. Calculate average time spent by platform and plot a bar chart.
5. Calculate average engagement by video category and plot a horizontal bar chart.
6. Plot time spent against engagement using a scatter plot.
7. Add titles, labels and layout adjustments, then display or save the figures.

SOURCE CODE

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_excel("socialmedia_dataset.xlsx")

# 1. Platform distribution
df["Platform"].value_counts().plot(kind="bar")
plt.title("Social Media Platform Distribution")
plt.xlabel("Platform")
plt.ylabel("Number of Records")
plt.tight_layout()
plt.show()

# 2. Average time spent by platform
df.groupby("Platform")["Total Time Spent"].mean().plot(kind="bar")
plt.title("Average Time Spent by Platform")
plt.xlabel("Platform")
plt.ylabel("Average Time Spent")
plt.tight_layout()
plt.show()

# 3. Average engagement by video category
(df.groupby("Video Category")["Engagement"].mean()
   .sort_values().plot(kind="barh"))
plt.title("Average Engagement by Video Category")
plt.xlabel("Average Engagement")
plt.ylabel("Video Category")
plt.tight_layout()
plt.show()

# 4. Scatter plot: time spent vs engagement
plt.scatter(df["Total Time Spent"], df["Engagement"])
plt.title("Time Spent vs Engagement")
plt.xlabel("Total Time Spent")
plt.ylabel("Engagement")
plt.tight_layout()
plt.show()
```

OUTPUT – VISUALIZATIONS

The following charts use a small illustrative sample to show the expected output format. Run the code above on the supplied project dataset to produce the actual charts and values.

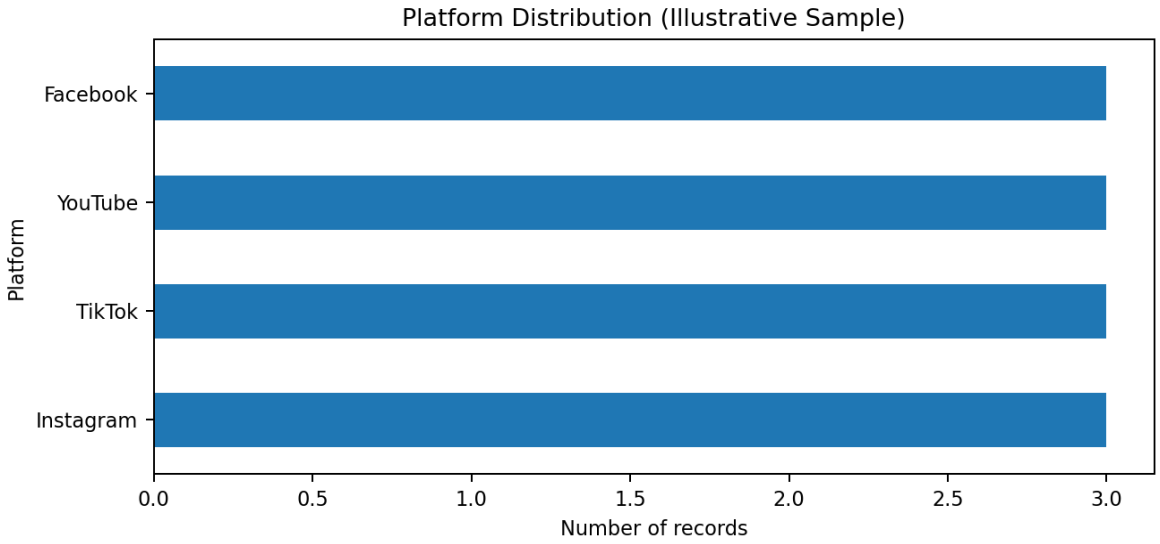


Figure 1. Platform distribution.

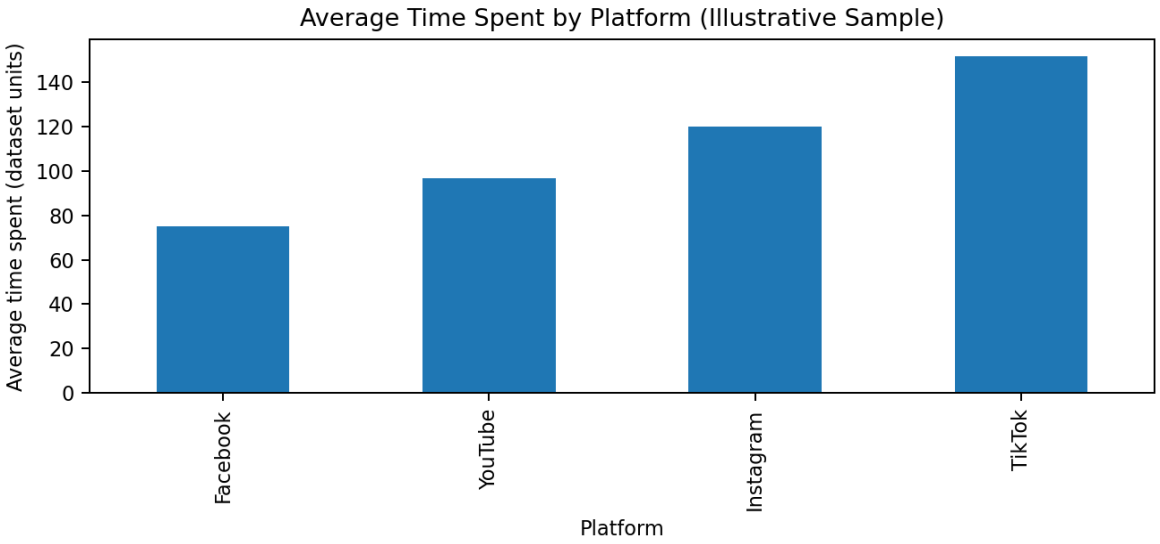


Figure 2. Average time spent by platform.

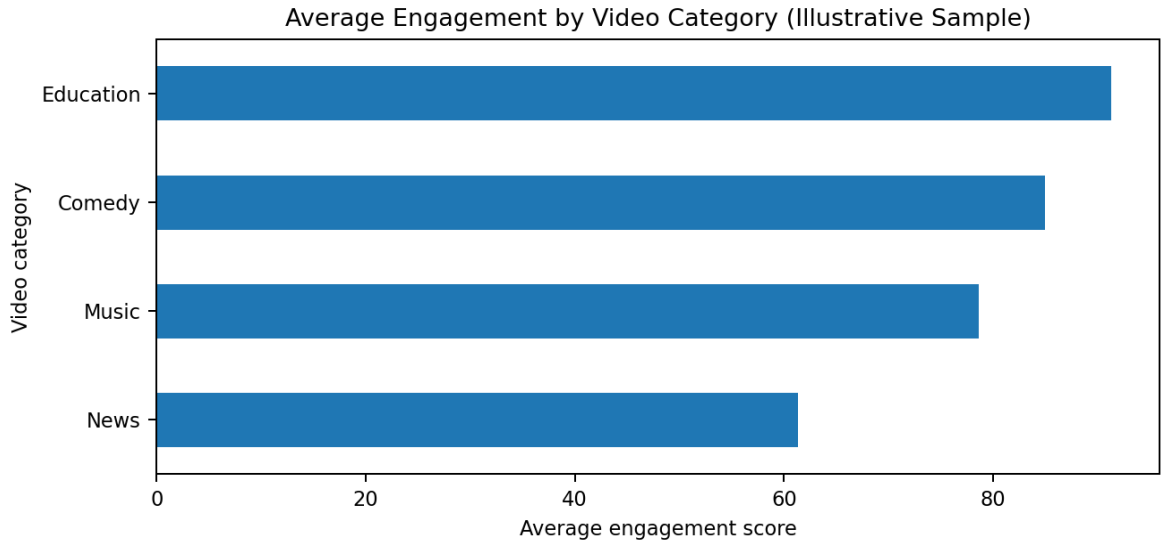


Figure 3. Average engagement by video category.

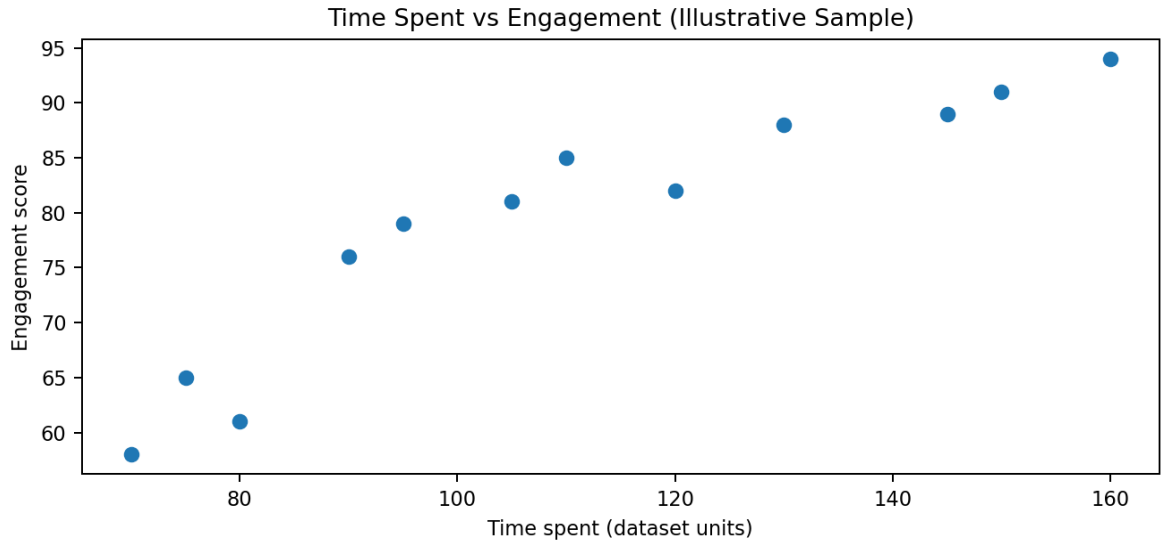


Figure 4. Relationship between time spent and engagement.

INTERPRETATION

Use the platform distribution chart to compare record counts, the average-time chart to compare usage values, and the category chart to compare typical engagement scores. The scatter plot shows whether larger time-spent values tend to occur with larger engagement values. A visible pattern is descriptive only and does not prove that time spent causes engagement.

RESULT

Different visualizations were created using Matplotlib, including bar charts, horizontal bar charts and a scatter plot. The charts present platform counts, platform-level time averages, category-level engagement averages and the relationship between time spent and engagement.